Experiment No: - 1.4

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1. Aim:

Write a program to find the **length of the longest substring without repeating characters** using the **Sliding Window optimization technique**.

2. Objective:

- To understand the concept of the sliding window technique.
- To find the longest substring containing unique characters.
- To avoid unnecessary repeated traversal of the string.
- To analyze the time and space complexity of the optimized approach.

3. Algorithm:

1. Take a string `s` as input.
2. Initialize:
   - `left = 0` to represent the starting point of the window.
   - `maxLength = 0` to store the maximum length.
   - An array/map to store the last position of each character.
3. Traverse the string using `right` from left to right.
4. If the current character has already appeared **inside the current window**, move `left` to one position after its previous occurrence.
5. Calculate the current window length:
   ```
   right - left + 1
   ```
6. Update `maxLength` if the current window is larger.
7. Print the longest substring length.